

STEP 1 — Install Libraries

```
!pip install transformers torch
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from transformers)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: Jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.2)
Requirement already satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from torch)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from Jinja2)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0)
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0)
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0)
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0)
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py)
```

STEP 2 — Import Libraries

```
from transformers import BartForConditionalGeneration, BartTokenizer
```

STEP 3 — Load Summarization Model

```
tokenizer = BartTokenizer.from_pretrained("facebook/bart-large-cnn")
model = BartForConditionalGeneration.from_pretrained("facebook/bart-large-cnn")

vocab.json:      899k/? [00:00<00:00, 31.7MB/s]
merges.txt:      456k/? [00:00<00:00, 19.4MB/s]

tokenizer.json:   1.36M/? [00:00<00:00, 52.4MB/s]

model.safetensors: 100% 1.63G/1.63G [00:26<00:00, 53.0MB/s]
Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights: 100% 511/511 [00:00<00:00, 825.31it/s, Materializing param=model.encoder.layers.11.self_attn_layer_norm
generation_config.json: 100% 363/363 [00:00<00:00, 17.2kB/s]
```

STEP 4 — Prepare Dataset (Different Text Types)

```

texts = [
    # News
    """Artificial Intelligence is rapidly transforming industries across the world.
    It helps in automation, decision-making, and improving efficiency.
    Many companies are investing heavily in AI technologies."""

    # Technical
    """Machine learning is a subset of artificial intelligence that focuses on building models
    that can learn from data. It includes supervised, unsupervised, and reinforcement learning te

    # Story
    """A young girl dreamed of becoming a scientist. Despite facing many challenges,
    she continued her studies and finally achieved her goal, inspiring many others."""

    # Blog
    """Healthy eating is important for maintaining good health. A balanced diet includes fruits,
    vegetables, proteins, and carbohydrates. Regular exercise also plays a key role."""

    # Conversation
    """Customer: I want to return my product.
    Agent: Please provide your order ID.
    Customer: My order ID is 56789.
    Agent: Thank you, we will process your request."""
]

```

STEP 5 — Generate Summaries

```

for i, text in enumerate(texts):
    inputs = tokenizer([text], max_length=1024, return_tensors='pt', truncation=True)
    summary_ids = model.generate(inputs['input_ids'], num_beams=4, max_length=60, min_length=20,
    summary_text = [tokenizer.decode(g, skip_special_tokens=True, clean_up_tokenization_spaces=False)

    print(f"\n--- Text {i+1} ---")
    print("Original:", text)
    print("Summary:", summary_text[0])

```

--- Text 1 ---
 Original: Artificial Intelligence is rapidly transforming industries across the world.
 It helps in automation, decision-making, and improving efficiency.
 Many companies are investing heavily in AI technologies.
 Summary: Artificial Intelligence is rapidly transforming industries. It helps in automation, decis

--- Text 2 ---
 Original: Machine learning is a subset of artificial intelligence that focuses on building models
 that can learn from data. It includes supervised, unsupervised, and reinforcement learning tec
 Summary: Machine learning is a subset of artificial intelligence that focuses on building models t

--- Text 3 ---
 Original: A young girl dreamed of becoming a scientist. Despite facing many challenges,
 she continued her studies and finally achieved her goal, inspiring many others.
 Summary: A young girl dreamed of becoming a scientist. Despite facing many challenges, she continu

--- Text 4 ---
 Original: Healthy eating is important for maintaining good health. A balanced diet includes fruits
 vegetables, proteins, and carbohydrates. Regular exercise also plays a key role.
 Summary: A balanced diet includes fruits, vegetables, proteins, and carbohydrates. Regular exercis

--- Text 5 ---
 Original: Customer: I want to return my product.
 Agent: Please provide your order ID.
 Customer: My order ID is 56789.
 Agent: Thank you, we will process your request.
 Summary: Customer: I want to return my product. Agent: Please provide your order ID. Customer: My

STEP 6 — Control Summary Length

```

text = texts[0]

# Short summary
inputs_short = tokenizer([text], max_length=1024, return_tensors='pt', truncation=True)
summary_ids_short = model.generate(inputs_short['input_ids'], num_beams=4, max_length=30, min_length=10)
short_summary_text = tokenizer.decode(summary_ids_short[0], skip_special_tokens=True, clean_up_tokenization_spaces=True)

# Medium summary
inputs_medium = tokenizer([text], max_length=1024, return_tensors='pt', truncation=True)
summary_ids_medium = model.generate(inputs_medium['input_ids'], num_beams=4, max_length=60, min_length=20)
medium_summary_text = tokenizer.decode(summary_ids_medium[0], skip_special_tokens=True, clean_up_tokenization_spaces=True)

# Long summary
inputs_long = tokenizer([text], max_length=1024, return_tensors='pt', truncation=True)
summary_ids_long = model.generate(inputs_long['input_ids'], num_beams=4, max_length=100, min_length=40)
long_summary_text = tokenizer.decode(summary_ids_long[0], skip_special_tokens=True, clean_up_tokenization_spaces=True)

print("\nShort Summary:\n", short_summary_text)
print("\nMedium Summary:\n", medium_summary_text)
print("\nLong Summary:\n", long_summary_text)

```

```

Short Summary:
Artificial Intelligence is rapidly transforming industries across the world. It helps in automati

Medium Summary:
Artificial Intelligence is rapidly transforming industries. It helps in automation, decision-maki

Long Summary:
Artificial Intelligence is rapidly transforming industries across the world. It helps in automati

```

STEP 7 — Test with Very Long Text

```

long_text = """Artificial Intelligence (AI) is growing rapidly and impacting many industries.
It is used in healthcare for diagnosis, in finance for fraud detection, and in education for pers
However, there are concerns about job loss and ethical issues. Governments are creating policies

inputs_long_text = tokenizer([long_text], max_length=1024, return_tensors='pt', truncation=True)
summary_ids_long_text = model.generate(inputs_long_text['input_ids'], num_beams=4, max_length=120, min_length=40)
summary_long_text = tokenizer.decode(summary_ids_long_text[0], skip_special_tokens=True, clean_up_tokenization_spaces=True)

print("\n--- Long Text Summary ---")
print(summary_long_text)

```

```

--- Long Text Summary ---
Artificial Intelligence (AI) is growing rapidly and impacting many industries. It is used in healt

```

STEP 8 — (Optional) Store Results

```

results = []

for text in texts:
    inputs = tokenizer([text], max_length=1024, return_tensors='pt', truncation=True)
    summary_ids = model.generate(inputs['input_ids'], num_beams=4, max_length=60, min_length=20, early_stopping=True)
    summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True, clean_up_tokenization_spaces=True)
    results.append(summary_text)

print(results)

```

Start coding or generate with AI.